VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
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Faculty of Computer Science and Engineering

Textbook Exercises: Performance

Advanced System Architectures

CO5059

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Hennessy & Patterson, Computer Architecture: A Quantitative Approach*, 5th ed., Chapter 1 case studies and exercises (1.1–1.18). A map of every exercise with its page and relevance to Lecture 1, worked solutions for the ones our lectures cover, and hints for the rest.*

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Contents

1 How to Use This File **1**

2 Exercise Map **1**

3 Worked Solutions: High Priority **2**
Exercise 1.1 2
Exercise 1.2 2
Exercise 1.3 2
Exercise 1.13 3
Exercise 1.14 3
Exercise 1.15 4
Exercise 1.16 4
Exercise 1.17 4

4 Worked Solutions: Medium Priority **5**
Exercise 1.18 5
Exercise 1.8 5
Exercise 1.6 5

5 Hints: Low Priority (beyond our Lecture 1) **6**

1 How to Use This File

- The exercise text is **not** copied here – open the book (`references/Computer Architecture - A Quantitative Approach.pdf`) at the page given. "Book p." is the printed page number; "PDF p." is the page in the PDF viewer.
- Each worked solution starts with a one-line summary of what the exercise asks, so you can check you are on the right problem.
- Chapter 1 of the 5th edition has no plain CPI/clock-cycle exercises; those are in Appendix A (A.1–A.4, see the ISA file) and Appendix C (C.2, C.3, C.7, see the pipelining file).

> **Note. Yield formula.** The 5th edition's text uses Yield $= 1/(1+D\cdot A)^N$ with $N \approx 13.5$ for a 40 nm process, but that makes the 130/90 nm chips of Exercises 1.1–1.3 have almost zero yield. The answers below use the classic formula from our lecture, Yield $= (1 + D \cdot A/4)^{-4}$, which is what these exercises were written for.

2 Exercise Map

Ex.	Book p.	PDF p.	Topic	Priority
1.1	62	91	Die yield (Power5); why defect rates differ	**High**
1.2	62	91	Dies per wafer, profit per wafer, which chip to make	**High**
1.3	62	91	Yield with a spare core; cost per good chip	**High**
1.4	63	92	Power supply sizing, disk idle power	Low (power)
1.5	63–64	92–93	Cooling doors and servers per rack	Low (power)
1.6	64	93	Discussion: what else matters besides benchmark results	Medium
1.7	64	93	Dual core at lower frequency/voltage; dynamic power	Low (power)
1.8	65	94	Moore's law projections, clock-rate limits	Medium
1.9	65	94	Energy saving for a real-time deadline	Low (power)
1.10	65–66	94–95	Server-farm power-saving options	Low (power)
1.11	66	95	FIT, MTTF, availability	Low (not covered)
1.12	66	95	MTTF of a server farm; cost of downtime	Low (not covered)
1.13	66	95	SPEC ratios, weighted time ratios, speedup	**High**
1.14	66–67	95–96	Amdahl's Law: vectorization	**High**
1.15	67	96	Amdahl's Law from a measured fraction of the *new* time	**High**
1.16	67–68	96–97	Amdahl's Law with a speedup and a slowdown	**High**
1.17	68	97	Amdahl's Law for parallel parts of two applications	**High**
1.18	68	97	Amdahl's Law with communication overhead	Medium

3 Worked Solutions: High Priority

Exercise 1.1 (book p. 62, PDF p. 91)

Asks: the yield of the IBM Power5 from the chip table (Fig. 1.22), and why its defect rate is lower than the other chips'.

Solution.

(a) $A = 389\ \text{mm}^2 = 3.89\ \text{cm}^2$, $D = 0.30/\text{cm}^2$:

$$\text{Yield} = \left(1 + \frac{0.30 \times 3.89}{4}\right)^{-4} = 1.292^{-4} \approx 0.36.$$

(b) It is made in an older, more mature 130 nm process. Defect density falls as a process matures (the learning curve), while the other two chips use a newer 90 nm process.

□

Exercise 1.2 (book p. 62, PDF p. 91)

Asks: profit per 300 mm wafer for a 150 mm^2 chip (\$20 profit per good chip) and a 250 mm^2 chip (\$25), both at the Power5's defect rate; which to build, and how to split 150 wafers/month given the demand.

Solution. Dies per wafer $= \pi(15)^2/A - \pi \cdot 30/\sqrt{2A}$, $D = 0.30$.

(a) $A = 1.5$: dies $= 471.2 - 54.4 = 416$; yield $= (1+0.1125)^{-4} = 0.653$; good ≈ 272; profit $\approx 272 \times \$20 \approx \$5{,}430$ per wafer.

(b) $A = 2.5$: dies $= 282.7 - 42.2 = 240$; yield $= (1+0.1875)^{-4} = 0.503$; good ≈ 121; profit $\approx 121 \times \$25 \approx \$3{,}020$ per wafer.

(c) The smaller chip: it earns $\approx 1.8\times$ more per wafer.

(d) Meeting the demand for the small chip needs $50{,}000/272 \approx 184$ wafers, more than the 150 available, and each of its wafers earns more than a large-chip wafer. So all 150 wafers go to the small chip ($\approx$ 40,700 chips). The book gives no profit figure for the Power5 itself, so that sub-question cannot be answered numerically.

□

Exercise 1.3 (book p. 62, PDF p. 91)

Asks: put a second core on the Opteron die and keep a chip if at least one core is good. Probability of a usable chip, and the new cost per chip if the old one cost \$20.

Solution.

(a) One core: $A = 1.99$, $D = 0.75$: $p = (1 + 0.75 \times 1.99/4)^{-4} = 1.373^{-4} \approx 0.28$. At most one of two cores defective:

$$p^2 + 2p(1-p) = 1 - (1-p)^2 = 1 - 0.72^2 \approx 0.48.$$

(b) The die doubles in area, so (ignoring edge effects) half as many dies fit per wafer and each die costs twice as much before yield. Cost per good chip $\approx \$20 \times 2 \times 0.28/0.48 \approx \$23.3$. The spare core *raises* the cost: the yield gain ($0.28 \to 0.48$) does not make up for doubling the area.

□

Exercise 1.13 (book p. 66, PDF p. 95)

Asks: choose between Opteron and Itanium 2 for a workload that is 60% wupwise-like, 20% ammp-like, 20% apsi-like, using the SPEC table (Fig. 1.17, book p. 43).

Solution. Data: execution time ratios Opteron/Itanium: wupwise 0.92, ammp 1.03, apsi 0.65 (times 51.5/56.1, 136/132, 150/231 s).

(a) By overall SPEC (geometric mean of SPECRatios) the Itanium 2 wins: 27.12 vs. 20.86, i.e. 1.30×.

(b) Weighted average of the time ratios: $0.6(0.92) + 0.2(1.03) + 0.2(0.65) = 0.89$.

(c) $1/0.89 \approx 1.13$: for this workload the Opteron is faster, the opposite of (a).

□

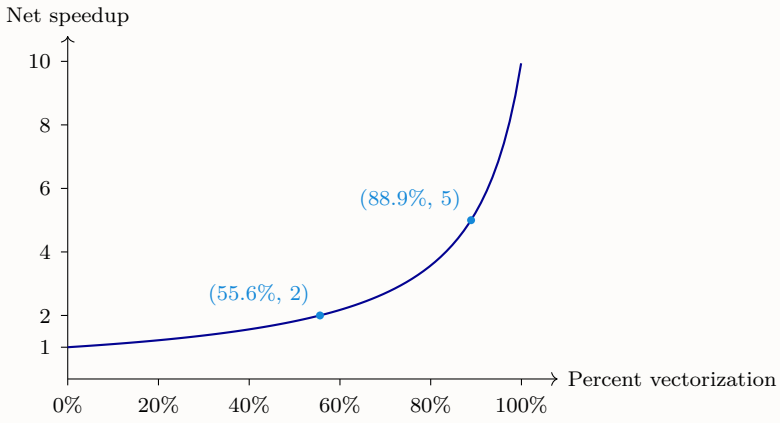
Remark. A more exact way is to weight the actual times: Opteron $0.6(51.5) + 0.2(136) + 0.2(150) = 88.1$ s, Itanium $0.6(56.1) + 0.2(132) + 0.2(231) = 106.3$ s, a speedup of 1.21. Either way the conclusion is the same: the right choice depends on *your* workload, not on the overall SPEC number.

Exercise 1.14 (book p. 66–67, PDF p. 95–96)

Asks: a vector mode 10× faster than normal mode; plot speedup vs. percent vectorization, and find the vectorization needed for various targets.

Solution. With vectorized fraction p: $\text{Speedup}(p) = 1/\big((1-p) + p/10\big)$.

(a) Speedup vs. percent vectorization:



The curve stays flat for a long time and shoots up only near 100%.

(b) $1/2 = 1 - 0.9p \Rightarrow p = 0.5/0.9 = 55.6\%$.

(c) New time = 0.5; vector part = $p/10 = 0.0556$; share = $0.0556/0.5 = 11.1\%$ of the *new* run time.

(d) Half of the maximum (10) is 5: $0.2 = 1 - 0.9p \Rightarrow p = 88.9\%$.

(e) At 70% with a 20× unit: $1/(0.3 + 0.7/20) = 1/0.335 = 2.99$. With the 10× unit the same speedup needs $0.335 = 1 - 0.9p \Rightarrow p = 73.9\%$. Raising vectorization by only ≈ 4 points matches doubling the vector hardware – probably the cheaper option.

□

Exercise 1.15 (book p. 67, PDF p. 96)

Asks: an enhancement is 10× faster and is in use for 50% of the *enhanced* run time. Overall speedup, and what fraction of the *original* time was converted.

Solution. Normalize the new time to 1: 0.5 unenhanced + 0.5 enhanced. The enhanced half would have taken $0.5 \times 10 = 5$ originally.

(a) Original time = $0.5 + 5 = 5.5$, so speedup = 5.5.

(b) $5/5.5 = 90.9\%$ of the original time used the fast mode. Check with Amdahl: $1/(0.091 + 0.909/10) = 5.5$. ✓

□

Remark. This is the classic trap: Amdahl's F is a fraction of the *original* time. Plugging in $F = 0.5$ would give only 1.82.

Exercise 1.16 (book p. 67–68, PDF p. 96–97)

Asks: a faster FP unit (2×, 20% of time) that slows data-cache accesses (1.5× slower, 10% of time). Overall speedup, and the new time shares.

Solution.

(a) $1/(0.8 + 0.2/2) = 1/0.9 = 1.11$.

(b) New time = $0.7 + 0.2/2 + 0.1 \times 1.5 = 0.7 + 0.1 + 0.15 = 0.95$; speedup = $1/0.95 = 1.053$.

(c) FP: $0.10/0.95 = 10.5\%$; data cache: $0.15/0.95 = 15.8\%$.

□

Exercise 1.17 (book p. 68, PDF p. 97)

Asks: two applications use 80% and 20% of a dual-core's resources; parallel parts get speedup 2. Speedup of each application alone, and of the whole system.

Solution.

(a) App 1, 40% parallel: $1/(0.6 + 0.4/2) = 1.25$.

(b) App 2, 99% parallel: $1/(0.01 + 0.99/2) = 1/0.505 = 1.98$.

(c) System, parallelizing app 1: $F = 0.8 \times 0.4 = 0.32$: $1/(0.68 + 0.16) = 1.19$.

(d) System, parallelizing app 2: $F = 0.2 \times 0.99 = 0.198$: $1/(0.802 + 0.099) = 1.11$.

Even though app 2 parallelizes almost perfectly, improving app 1 helps the system more – it is the common case. □

4 Worked Solutions: Medium Priority

Exercise 1.18 (book p. 68, PDF p. 97)

Asks: speedup on N processors for an 80%-parallel program, then with communication overhead that grows per processor or per doubling, and the best N.

Solution.

(a) $1/(0.2 + 0.8/N)$.

(b) Counting 7 added processors: overhead $= 7 \times 0.5\% = 3.5\%$: $1/(0.2+0.1+0.035) = 2.99$ (2.94 if you charge all 8 processors).

(c) 1→8 is three doublings, overhead 1.5%: $1/(0.2 + 0.1 + 0.015) = 3.17$.

(d) $1/\big(0.2 + 0.8/N + 0.005 \log_2 N\big)$.

(e) Time $T(N) = (1 - P) + P/N + 0.005 \log_2 N$. Setting $dT/dN = -P/N^2 + 0.005/(N \ln 2) = 0$ gives $N^* = P/(0.005 \ln 2) \approx 289P$ (about 231 processors for $P = 0.8$).

□

Exercise 1.8 (book p. 65, PDF p. 94)

Asks: project transistor counts and clock rates ten years ahead, and explain what limited clock growth.

Solution. (a) Doubling every 2 years for 10 years: $2^5 = 32\times$ (about 100× with an 18-month doubling). (b)/(c) Apply the 1990s growth rate and the recent much lower rate from Section 1.4/Fig. 1.11 to the 2005 clock rate. (d) Power (the power wall): higher clocks need more power than can be cooled. The extra transistors now go into multiple cores and larger caches. (e) Apply the stated slowing trend; growth keeps decreasing (roughly halving again, to the order of 10–20%/year). □

Exercise 1.6 (book p. 64, PDF p. 93)

Asks: why someone might choose the server that loses on the reported power/performance benchmarks.

Solution. Reported benchmarks may not match your workload (single-thread performance, floating point, memory size and bandwidth, I/O); also price and price-performance, software compatibility, reliability/availability, support, and the fact that a vendor chose which results to report. □

5 Hints: Low Priority (beyond our Lecture 1)

These use the dynamic power equation and dependability metrics from Sections 1.5 and 1.7 of the book, which our lecture only mentioned.

- **1.4, 1.5** (pp. 63–64): add the component powers from Fig. 1.23, divide by the supply efficiency; servers per cooling door = door capacity / power per server.
- **1.7, 1.9, 1.10** (pp. 64–66): dynamic power $\propto \frac{1}{2}CV^2f$; energy $\propto CV^2$. Halving V and f cuts energy to 1/4 and power to 1/8.
- **1.11** (p. 66): MTTF = 10^9/FIT hours (10^7 h for FIT 100); availability = MTTF/(MTTF + MTTR); with 1000 parts that all fail together, MTTF/1000.
- **1.12** (p. 66): treat failures as independent with rate 1/MTTF per machine; compare the extra hardware cost with the downtime cost from Fig. 1.3.